UNIVERSITÀ DEGLI STUDI DI MILANO

FACOLTÀ DI SCIENZE POLITICHE, ECONOMICHE E SOCIALI

Convolutional Neural Network Architectures for Rock–Paper–Scissors Image Classification

End-Of-Course Project

Leila Cielok

Data Science for Economics (Master's Degree)
II Year

I declare that this material, which I now submit for assessment, is entirely my own work and has not been taken from the work of others, save and to the extent that such work has been cited and acknowledged within the text of my work. I understand that plagiarism, collusion, and copying are grave and serious offences in the university and accept the penalties that would be imposed should I engage in plagiarism, collusion or copying. This assignment, or any part of it, has not been previously submitted by me or any other person for assessment on this or any other course of study.

February 28, 2026

Abstract

This project investigates the impact of architectural complexity on the performance and robustness of convolutional neural networks for image classification, using the Rock–Paper–Scissors hand gesture task as a controlled case study. Three CNN architectures with progressively increasing capacity are designed and compared under identical training conditions, including consistent data preprocessing, augmentation, and optimization strategies. Model selection is guided by a constrained hyperparameter tuning procedure based on validation accuracy. Beyond standard in-distribution evaluation, particular emphasis is placed on generalization under domain shift through testing on an external dataset of unconstrained, real-world images. The experimental results reveal a clear relationship between model capacity and in-distribution performance, while also highlighting that strong validation accuracy does not necessarily guarantee robustness to distributional changes. The findings underscore the importance of evaluating neural network models beyond held-out validation data when assessing their reliability in real-world deployment scenarios.

1 Introduction

Convolutional Neural Networks (CNNs) have become a standard approach for image classification tasks due to their ability to automatically learn hierarchical feature representations from raw visual data. In recent years, CNN-based models have demonstrated strong performance across a wide range of computer vision applications, from object recognition to gesture and action classification.

Despite these successes, the relationship between architectural complexity and model generalization remains a central issue in deep learning. While increasing model capacity often leads to improved performance on held-out validation data, it does not necessarily translate into robust behaviour when models are exposed to distributional shifts or unconstrained real-world inputs. This challenge is particularly relevant for vision tasks involving variability in lighting, backgrounds, and acquisition conditions, such as hand gesture recognition.

Within this context, the present study uses the Rock–Paper–Scissors image classification task as a controlled benchmark to analyse how different CNN architectures behave as model complexity increases. By comparing multiple architectures under a consistent experimental protocol, the project aims to provide insight into how architectural choices affect learning dynamics, error structure, and generalization properties under both standard and shifted data distributions.

The remainder of the report is organized as follows. Section 2 describes the dataset and preprocessing pipeline, Section 3 presents the model architectures, Section 4 reports the experimental results, and Section 5 discusses generalization under domain shift.

2 Data Exploration and Preprocessing

2.1 Dataset Description and Exploratory Analysis

The dataset used in this project was obtained from a publicly available Kaggle repository (`https://www.kaggle.com/datasets/drgfreeman/rockpaperscissors`) containing images of hand gestures for the Rock–Paper–Scissors game. The dataset consists of a total of 2,188 RGB images, evenly distributed across the three target classes: rock (726 images), paper (710 images), and scissors (752 images). All images were captured against a green background under relatively consistent lighting and white balance conditions, resulting in a visually homogeneous and well-curated dataset.

Each image is stored in PNG format with an original resolution of 300 $\times$ 200 pixels and is organized into class-specific subdirectories, enabling a straightforward mapping between file structure and class labels. An initial exploratory analysis confirmed the near-balanced class distribution and the absence of missing or corrupted files. Random visual inspection of samples from each class further revealed that the hand gestures are generally well-centered and clearly distinguishable, with limited intra-class variability compared to real-world acquisition scenarios. This level of consistency makes the dataset particularly suitable for controlled experiments focused on architectural comparisons rather than robustness to extreme visual noise.

To avoid any form of data leakage, external datasets are introduced only at a later stage of the analysis. These datasets are used exclusively for evaluation and play no role in model training or model selection, allowing generalization to be assessed independently of the training distribution.

2.2 Train – Validation Split

The dataset was split into training and validation subsets using a stratified sampling strategy, with 80% of the images allocated to training and 20% reserved for validation. Stratification was performed at the file level to preserve the original class proportions in both splits and to guarantee that all classes are represented in each subset.

A fixed random seed was used to ensure reproducibility of the split across runs. The validation set was held out entirely from training and used exclusively for model selection, hyperparameter tuning, and performance evaluation. This splitting strategy provides a reliable estimate of in-distribution generalization while maintaining methodological transparency and repeatability.

2.3 Image Preprocessing: Resizing and Normalization

After splitting the dataset into training and validation subsets, a common preprocessing pipeline was applied to all images: to ensure compatibility with convolutional neural network architectures and to standardize the input space, images were resized to a fixed square resolution of 96 × 96 pixels. Resizing was performed using bilinear interpolation with antialiasing, preserving the essential spatial structure of the hand gestures while reducing computational cost and memory requirements.

Following resizing, pixel intensities were converted to floating-point format and normalized to the [0, 1] range, with the aim of facilitating stable and efficient optimization by preventing large gradient magnitudes during training. This normalization step was applied directly within the data pipeline rather than as a dedicated rescaling layer inside the models, ensuring consistent preprocessing across all architectures and simplifying model definitions.

2.4 Data Augmentation

Although the dataset is visually consistent and relatively clean, light data augmentation was applied during training to encourage better generalization and to reduce sensitivity to small spatial variations. Augmentation operations were deliberately kept mild to avoid introducing unrealistic distortions, given the already constrained nature of the dataset.

Specifically, the training pipeline includes random horizontal flips, small rotations, zooming, translations, and slight contrast adjustments, applied exclusively to the training set to ensure that validation data remains untouched and representative of the original distribution. These transformations are applied on-the-fly at batch loading time, meaning that each epoch exposes the model to a different randomized version of the same training samples. This strategy allows the models to learn more robust feature representations while preserving a fair and unbiased evaluation protocol.

3 CNN Architecture and Training

3.1 Design of CNN Architectures with Incremental Complexity

The three convolutional neural network architectures evaluated in this study were designed to form a controlled complexity hierarchy, in which representational capacity, depth, and regularization mechanisms are progressively increased while keeping the input resolution and training protocol fixed.

Formally, each model defines a mapping $f(\mathbf{x}_0; \Theta)$ from an input image $\mathbf{x}_0 \in \mathbb{R}^{H \times W \times 3}$ to class probabilities via a softmax layer.

To make the concept of incremental increase in model capacity more explicit, Table 1 reports the number of trainable parameters and effective model size for each architecture.

Table 1: *Number of trainable parameters and model size across architectures.*

Model	Trainable parameters	Model size (trainable)
Model A	2,814	11 KB
Model B	249,331	974 KB
Model C	45,523	178 KB

Model A. Model A is a deliberately lightweight architecture intended as a minimal baseline. It relies on a small number of depthwise-separable convolutional layers followed by max-pooling operations:

$$\mathbf{x}_{l+1} = \sigma(\texttt{SepConv}(\mathbf{x}_l)), \qquad \mathbf{x}_{l+2} = \texttt{MaxPool}(\mathbf{x}_{l+1})$$

After three such stages, global average pooling (GAP) is applied:

$$\mathbf{z} = \texttt{GAP}(\mathbf{x}_L),$$

followed by a small dense head with dropout:

$$\hat{\mathbf{y}} = \texttt{Softmax}(W_2\, \texttt{ReLU}(W_1 \mathbf{z})).$$

The use of separable convolutions significantly reduces the number of trainable parameters and computational cost (2,814 trainable parameters; see Table 1), enforcing a strong inductive bias toward learning coarse spatial patterns. The adoption of global average pooling further limits the model's capacity to memorize spatially localized features. As a result, model A primarily captures global shape cues rather than fine-grained gesture details.

Model B. Model B represents a modest increase in expressiveness and corresponds to a conventional shallow CNN. It employs standard convolutional layers interleaved with max-pooling

$$\mathbf{x}_{l+1} = \sigma(\texttt{Conv2D}(\mathbf{x}_l))\,, \qquad \mathbf{x}_{l+2} = \texttt{MaxPool}(\mathbf{x}_{l+1}).$$

After two convolution–pooling stages, spatial feature maps are flattened:

$$\mathbf{z} = \texttt{Flatten}(\mathbf{x}_L),$$

and passed through a dense classifier:

$$\hat{\mathbf{y}} = \texttt{Softmax}(W_2\,\texttt{ReLU}(W_1\mathbf{z})).$$

Compared to model A, this design substantially increases parameter count (249,331 trainable parameters) and spatial flexibility, but reduces the implicit spatial regularization induced by global pooling. This makes model B more expressive but also more prone to sensitivity under distributional shifts, as later evidenced in the evaluation on the custom dataset.

Model C. Model C is a deeper architecture optimized for stable training and robust feature extraction under constrained data regimes. With 45,523 trainable parameters, it achieves a favorable capacity–efficiency trade-off. The architecture starts with a strided convolutional stem:

$$\mathbf{x}_1 = \texttt{Conv}_{s=2}(\mathbf{x}_0),$$

followed by repeated blocks composed of depthwise separable convolutions with layer normalization. Let $F(\mathbf{x};\theta)$ denote the composite transformation implemented by a depthwise separable block including convolution, normalization, and nonlinearity. Residual connections are conditionally applied as:

$$\mathbf{y} = \begin{cases} F(\mathbf{x};\theta) + \mathbf{x}, & \text{if } s = 1 \text{ and } C_{\text{in}} = C_{\text{out}}, \\ F(\mathbf{x};\theta), & \text{otherwise.} \end{cases}$$

Unlike classical ResNet architectures, no projection shortcut is used when dimensions change, keeping the block lightweight while preserving identity-based gradient flow whenever tensor shapes match.

Spatial resolution is reduced progressively through strided convolutions rather than pooling, leading to feature maps at resolutions 48×48, 24×24, and 12×12. After staged processing, global average pooling aggregates spatial information:

$$\mathbf{z} = \texttt{GAP}(\mathbf{x}_L),$$

followed by a compact dense head with dropout:

$$\hat{\mathbf{y}} = \texttt{Softmax}(W_2\,\texttt{Dropout}(\texttt{ReLU}(W_1\mathbf{z}))).$$

3.2 Architectural Design Choices and Technical Rationale

While the previous subsection described the structural composition of each architecture, we now discuss the underlying design rationale and regularization choices that ensure training stability and fair comparison.

First, depthwise separable convolutions, already introduced in Models A and C, are primarily motivated by robustness under limited data conditions. In Model C, separable convolutions are implemented explicitly via depthwise and pointwise operations and are combined with conditional residual connections. These residual shortcuts are applied only when spatial resolution and channel dimensionality match, improving gradient flow without introducing projection parameters.

Second, Layer Normalization (LN) is employed instead of Batch Normalization in Model C. Since training is performed with relatively small batch sizes and with data augmentation enabled, LN provides more stable normalization statistics that are independent of batch composition. This choice improves convergence consistency and reduces sensitivity to batch-size variations.

Beyond their structural description, the downsampling mechanisms differ systematically across architectures. Models A and B rely on MaxPooling layers for progressive spatial reduction, whereas Model C performs downsampling through strided convolutions (both in the initial stem and in depthwise blocks).

This design allows Model C to learn feature extraction and resolution reduction jointly, preserving richer intermediate representations.

Consistent with their parameter-efficient design, Models A and C employ Global Average Pooling at the classification stage. In contrast, Model B uses flattening, resulting in a higher-capacity fully connected head.

Label smoothing is applied uniformly across all models through the categorical cross-entropy loss, preventing overconfident predictions and improving robustness under domain shift. In Model C, the optional initialization of output biases using empirical log-prior class probabilities further stabilizes early training dynamics, particularly in the presence of mild class imbalance.

Across all architectures, optimization is performed using the Adam optimizer with default β_1 and β_2 coefficients. Model A is trained with a learning rate of 10^{-3}, Model B uses the default Adam learning rate, and Model C adopts a reduced learning rate of 3×10^{-4} to improve optimization stability in the deeper architecture.

Training is regularized through early stopping monitored on validation accuracy with restoration of the best-performing weights and an adaptive learning-rate schedule (ReduceLROnPlateau), which reduces the learning rate when validation performance fails to improve for a predefined number of epochs. Together, these mechanisms promote stable convergence and mitigate overfitting while limiting unnecessary computation once validation performance saturates.

3.3 Hyperparameter Tuning and Final Model Selection

Hyperparameter tuning is carried out through a constrained grid search applied consistently across all three architectures (Models A, B, and C), exploring a small set of high-impact training hyperparameters: learning rate, batch size, and the use of data augmentation. For each architecture, all combinations in the grid are trained using the same stratified train–validation split, and model selection is based on validation accuracy computed on the held-out validation set. Early stopping is employed during each run to prevent overfitting and to limit unnecessary computation once validation performance plateaued, enabling an efficient yet systematic exploration of the search space. All trials are logged in a dedicated results table (`tuning_results.csv`), ensuring full reproducibility and allowing direct comparison across architectures under matched tuning conditions. The best configurations identified for each model are summarized in Table 2.

Table 2: *Best tuning configurations per model, reporting the best run with and without augmentation.*

Model	Learning rate	Batch size	Augment	Val. accuracy
A	1×10^{-3}	32	No	0.6461
A	1×10^{-3}	32	Yes	0.6644
B	3×10^{-4}	32	No	0.9658
B	1×10^{-3}	32	Yes	0.9658
C	3×10^{-4}	16	No	0.9954
C	5×10^{-4}	32	Yes	0.9680

The best configuration identified for each architecture is subsequently used for a clean final training run with checkpointing and evaluation artefacts generated accordingly.

Section 3 presents a detailed comparison of validation performance across architectures and the resulting model ranking. The overall best-performing model is then selected for external generalization tests on the two additional datasets.

4 Evaluation and Analysis

Building on the architectural comparison introduced in Section 2, this section evaluates how increasing model complexity translates into empirical performance differences.

This section presents the results obtained from the training and evaluation pipeline, focusing on the outputs that are most informative for assessing model performance, stability, and generalization. Although the experimental framework produces a broad set of diagnostic artifacts during development, only a selected subset is reported here in order to avoid redundancy and to emphasize interpretability.

In particular, the results include:

i. a quantitative comparison across architectures based on validation performance, including accuracy, precision, recall, and F1-score, used to support model selection (Section 4.1);

ii. an analysis of training dynamics through loss and accuracy curves on both training and validation sets, providing insight into convergence behaviour and learning stability (Section 4.2);

iii. a qualitative analysis of misclassified validation examples, aimed at identifying systematic error patterns and model-specific failure modes (Section 4.3);

iv. an integrated assessment of overfitting and underfitting, based on performance metrics, learning curves, and error structure (Section 4.4).

Additional intermediate diagnostics generated during model development were used exclusively for debugging and validation purposes and are not included here, as they do not provide additional insight beyond the selected evaluation outputs presented above.

4.1 Performance Metrics

Model performance was assessed using standard multi-class classification metrics, including accuracy, precision, recall, and F1-score, computed on a held-out validation set. Each metric captures a different and complementary aspect of predictive behaviour, allowing a comprehensive view of both predictive quality and learning behaviour and a nuanced comparison across architectures beyond accuracy alone. While accuracy summarizes overall correctness, precision and recall capture complementary aspects of class-specific performance: precision reflects the model's ability to avoid false positive assignments, whereas recall measures its capacity to correctly retrieve all instances of a given class. The F1-score balances these two dimensions and is particularly informative when comparing models with different error profiles. Support values confirm that all metrics are computed on a nearly balanced validation set, ensuring that macro-averaged scores are meaningful and not driven by class imbalance.

Table 3: *Validation performance of the best tuned configuration for each architecture. Metrics are macro-averaged over classes (3 decimal precision).*

Model	Accuracy	Precision (macro)	Recall (macro)	F1-score (macro)
Model A (best)	0.651	0.651	0.650	0.645
Model B (best)	0.961	0.962	0.961	0.961
Model C (best)	0.993	0.993	0.993	0.993

As shown in Table 3, a clear and substantial improvement emerges across architectures, with a pronounced performance gap between the baseline model and the two more expressive networks.

The baseline architecture (Model A) achieves a validation accuracy of 0.651 and a macro F1-score of 0.645, indicating that it fails to reliably separate the three gesture classes even under in-distribution conditions. The close alignment between macro precision (0.651) and recall (0.650) suggests that performance limitations are not driven by a specific bias toward false positives or false negatives, but rather by an overall inability to extract sufficiently discriminative representations. In practical terms, this behaviour corresponds to frequent class confusion and unstable decision boundaries, consistent with limited representational capacity.

Model B marks a decisive qualitative shift in performance. Validation accuracy rises sharply to 0.961, with macro precision, recall, and F1-score all exceeding 0.96. This substantial gain relative to Model A indicates that the architectural modifications introduced in Model B enable the learning of far more expressive and class-separable features. The near-perfect balance between precision and recall across classes suggests that errors are both rare and symmetrically distributed, pointing to stable and well-calibrated decision regions.

The most expressive architecture (Model C) further consolidates these gains, achieving a validation accuracy of 0.993 and macro F1-score of 0.993. The marginal improvement over Model B reflects a transition from strong to near-saturated performance within the validation distribution. At this level, residual misclassifications become sparse and largely non-systematic. Importantly, the concurrent improvement in all reported metrics indicates that performance gains are not the result of a precision–recall trade-off, but rather of a genuine and uniform reduction in classification errors.

Taken together, these results reveal a two-stage progression in model capability. The transition from Model A to Model B produces a dramatic improvement driven by increased representational power, while the transition from Model B to Model C yields a more incremental refinement toward near-complete class separability. This progression provides a principled and empirically grounded justification for selecting Model C as the final architecture for subsequent generalization and robustness analyses.

Prediction confidence on the validation set. In addition to accuracy-based metrics, the distribution of predicted probabilities provides a coarse proxy for predictive confidence. Although this does not constitute a formal calibration assessment, it offers a simple summary of how confident each model is in its predictions. On the validation set, the mean maximum softmax probability increases markedly with model expressiveness (Model A: 0.44, Model B: 0.90, Model C: 0.95). This trend mirrors the improvement observed in classification accuracy and indicates that higher-performing architectures not only make fewer errors but also assign substantially higher confidence to their decisions.

The relatively low average confidence of Model A is consistent with its limited discriminative capacity and frequent class confusion, whereas Models B and C operate in a high-confidence regime with reduced uncertainty. Importantly, class-averaged predicted probabilities remain approximately balanced across classes, suggesting that improvements are not driven by degenerate class priors but by genuinely more separable representations. For completeness, validation-set confidence diagnostics are summarized in Table 4.

Table 4: *Prediction confidence diagnostics on the validation set: mean and standard deviation of the maximum softmax probability across samples.*

Model	Val max-prob ($\mu \pm \sigma$)
Model A (best)	0.44 ± 0.09
Model B (best)	0.90 ± 0.14
Model C (best)	0.95 ± 0.06

4.2 Training Dynamics

Training dynamics were analysed by jointly examining the evolution of loss and accuracy on both the training and validation sets. Figure 1 provides insight into how each architecture learns over time, whether performance improvements arise from genuine feature learning or from overfitting, and how effectively optimization translates into generalization within the validation distribution.

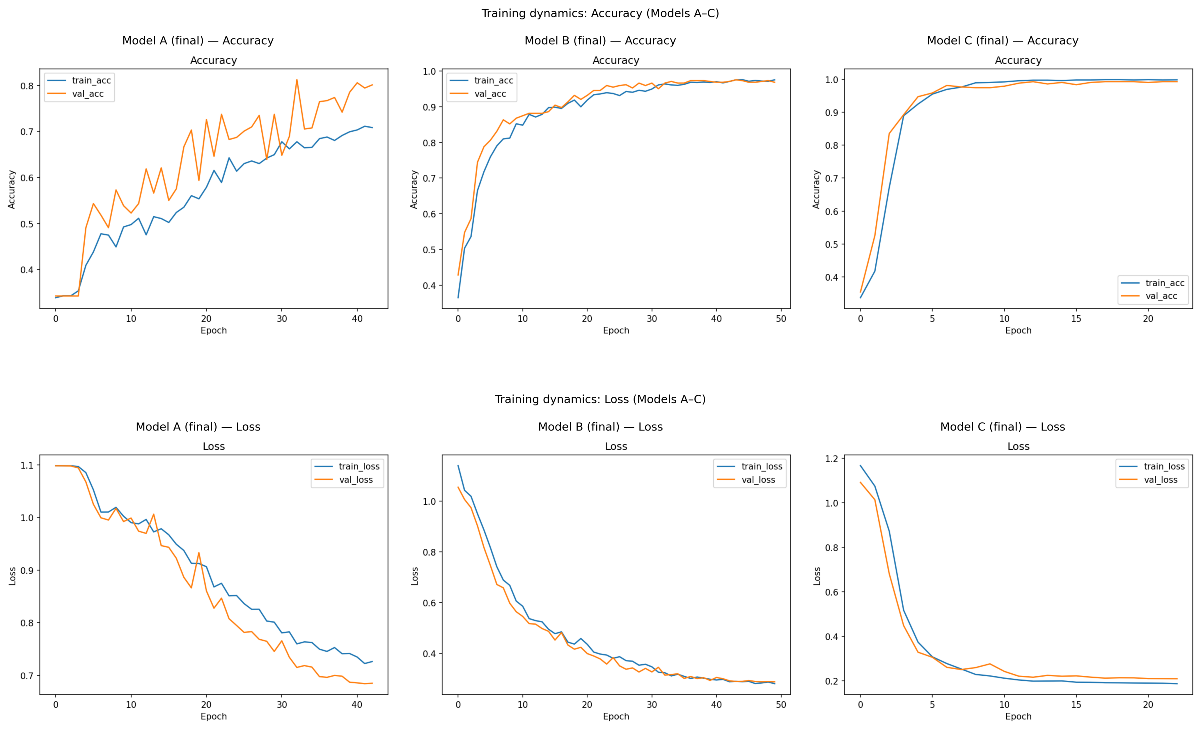

Figure 1: *Training and validation accuracy (top) and loss (bottom) curves for the three final architectures.*

For the baseline architecture A, both training and validation loss decrease steadily and approach a plateau; accuracy improves slowly and remains substantially below models B/C. Validation accuracy is noticeably noisier and often slightly higher than training accuracy, consistent with a limited-capacity model trained under augmentation/ regularization rather than with memorization-driven overfitting. The absence of a widening gap in the overfitting direction (i.e., training performance substantially exceeding validation performance) indicates that the model does not overfit the data. Instead, the early stagnation of performance suggests limited representational capacity: the model is unable to further reduce classification error even on the training set. In practical terms, this behaviour is characteristic of mild underfitting, where the architecture captures only coarse visual patterns and fails to exploit additional training iterations to improve class discrimination.

Model B exhibits a markedly different learning trajectory. Loss decreases steadily across epochs for both training and validation sets, accompanied by a rapid and sustained increase in accuracy. The close alignment between training and validation curves throughout the optimization process indicates stable learning and efficient use of model capacity. From a practical perspective, this pattern suggests that the architectural enhancements introduced in model B enable the network to learn more expressive features without sacrificing generalization, leading to consistent improvements in both error reduction and predictive accuracy.

Model C shows similarly smooth and stable convergence, but with more rapid early error reduction and lower final loss values. Training and validation curves remain tightly coupled, with only a minimal and non-systematic generalization gap (the training loss appears slightly below validation loss at convergence). This pattern indicates that the increased model capacity does not translate into late-stage divergence between training and validation performance. Instead, the adopted training configuration appears sufficient to maintain stable generalization within the validation distribution. The model is therefore able to leverage its higher representational capacity without exhibiting clear signs of memorization-driven overfitting. In practical terms, this behaviour corresponds to rapid acquisition of highly discriminative features and convergence toward a stable low-loss regime rather than unstable or capacity-induced degradation.

Overall, the progression of training dynamics across models mirrors the performance trends observed in the classification metrics. Model A is constrained by insufficient capacity, model B achieves an effective balance between learning and generalization, and model C combines high expressiveness with stable optimization. These dynamics strongly suggest that the observed performance differences are rooted in structural properties of the architectures rather than in optimization artefacts or training instability.

4.3 Analysis of Misclassified Examples

When informative, a qualitative analysis of misclassified examples was conducted to identify systematic error patterns and to complement the quantitative evaluation provided by aggregate performance metrics. In this context, *misclassified samples* are defined as validation images for which the predicted class label differs from the ground-truth label. For each such instance, the predicted class confidence is computed as the maximum softmax probability associated with the model's output. This allows misclassifications to be inspected not only in terms of correctness, but also in terms of model confidence.

To provide a compact overview of class-level error patterns before moving to instance-level inspection, Figure 2 reports the validation confusion matrices for the three final architectures. These matrices summarize how misclassifications are distributed across classes and offer a global complement to the analysis presented below.

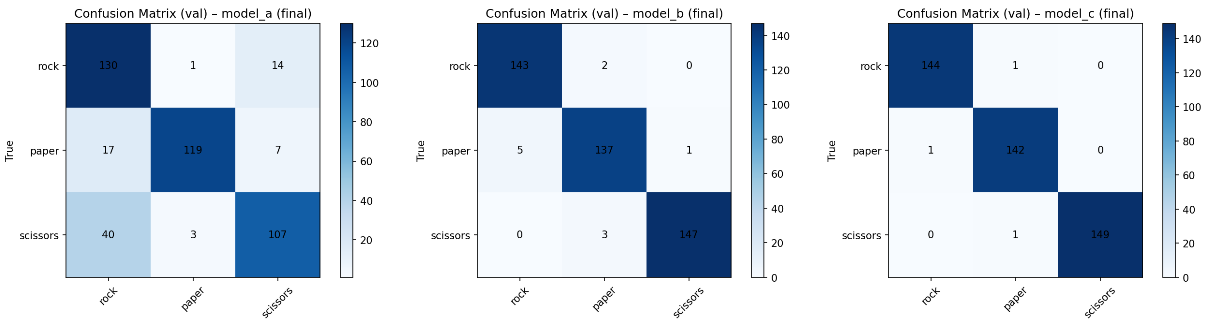

Figure 2: *Validation confusion matrices for the three final architectures (models A–C).*

Misclassified samples are ranked according to their prediction confidence and a fixed number of the most confident errors is selected for qualitative inspection. This design choice focuses the analysis on the most informative failure cases, namely those in which the model assigns high confidence to an incorrect prediction. Such errors are particularly relevant, as they reveal systematic confusions in the learned representation rather than borderline or noisy instances. Representative misclassified examples for each final architecture are shown in the Appendix, Figures 3, 4 and 5, ordered by decreasing prediction confidence.

The qualitative patterns observed differ substantially across architectures.

For Model A, misclassifications are frequent but not uniformly distributed across class pairs. The confusion matrix indicates that the dominant error mode is confusing scissors with rock (40 instances), with additional leakage of paper into rock and rock into scissors. Interestingly, the grid of most confident misclassifications highlights a different aspect: among the highest-confidence errors, several samples are predicted as paper with moderate confidence (around 0.6–0.8). This does not imply that paper is the most frequent erroneous prediction, but rather that a subset of paper-type false positives is associated with higher confidence than the more common scissors-to-rock confusions. Overall, this pattern supports the underfitting diagnosis: the model relies on coarse global cues and fails to establish consistently sharp decision boundaries between visually similar hand configurations.

For Model B, misclassifications become substantially fewer and more localized. The confusion matrix shows only a small number of errors per class, with no dominant confusion pattern. The grid of most confident errors reveals that mistakes typically arise in visually ambiguous instances, such as transitional hand poses or subtle finger extensions. Although a few errors appear with relatively high confidence (e.g., confident confusion between paper and rock), these cases are rare and do not form a systematic pattern. Compared to Model A, the internal representation appears more stable and class-separable, as reflected in both the sharp reduction in off-diagonal confusion counts and the dramatically increased overall accuracy.

For Model C, misclassifications are extremely rare and highly localized. The confusion matrix is nearly diagonal, with only isolated off-diagonal entries. The few errors observed in the grid correspond to borderline or visually ambiguous cases, such as transitional poses or minor deviations from canonical gesture shapes. Even when confidence values are high, these errors do not reveal a consistent structural bias toward a specific incorrect class. Rather, they appear to stem from intrinsic visual ambiguity in individual samples. This qualitative pattern aligns with the near-saturated performance metrics and supports the conclusion that residual errors are driven primarily by signal-level ambiguity rather than representational limitations.

In light of the above, the qualitative analysis of misclassified examples reinforces the conclusions drawn from performance metrics and training dynamics. As model capacity increases, errors transition from frequent and structurally biased (Model A), to sparse and ambiguity-driven (Models B and C). This progression indicates that improvements in classification performance are accompanied by a qualitative shift in the nature of errors, from representational insufficiency to near-irreducible visual ambiguity.

4.4 Overfitting and Underfitting Assessment

The presence or absence of overfitting and underfitting was assessed by jointly analysing quantitative performance metrics, training and validation learning curves, and the qualitative structure of classification errors. This combined perspective allows potential optimization artefacts to be distinguished from genuine limitations in model capacity.

For the baseline architecture (Model A), clear evidence of underfitting emerges. Training accuracy saturates at relatively low levels and does not substantially improve with additional epochs, while training and validation loss curves remain closely aligned throughout optimization. This behaviour indicates that the model fails to reduce error even on the training data, effectively ruling out memorization-driven overfitting and instead pointing to insufficient representational capacity. The qualitative inspection of misclassified examples further supports this diagnosis: errors are frequent and structurally biased (notably the systematic confusion of scissors with rock), revealing coarse decision boundaries and incomplete feature learning rather than noise sensitivity.

In contrast, no evidence of overfitting is observed for the more expressive architectures. For both models B and C, training and validation accuracy curves evolve in parallel, and validation loss does not exhibit late-stage divergence. The generalization gap remains small and stable, indicating that performance improvements are not driven by memorization of the training set but by the learning of increasingly discriminative representations.

Model B operates in a regime of effective capacity: it substantially reduces classification error while maintaining stable convergence and a balanced error profile across classes. Model C further increases representational power without inducing instability or widening the generalization gap. Although its performance approaches saturation on the validation set, the absence of divergence between training and validation metrics suggests that the model remains well-regularized under in-distribution conditions rather than over-parameterized.

Taken together, these findings indicate a progressive transition from capacity-limited underfitting (Model A) to effective high-capacity learning (Models B and C). Importantly, improvements in accuracy are not accompanied by the characteristic signatures of overfitting, supporting the conclusion that architectural expressiveness, rather than training instability, explains the observed performance differences.

5 Generalization Test

5.1 External In-Distribution Evaluation

In order to disentangle generalization within the same acquisition domain from robustness to genuine domain shift, the best model (model C) was additionally evaluated on the *rps-cv-images* dataset. This dataset is external to the training and validation splits, but shares similar characteristics in terms of background, lighting conditions, and gesture presentation.

The model achieves near-perfect performance on this benchmark, with an overall accuracy exceeding 99% across 2,188 test images and only a handful of isolated misclassifications. The confusion matrix is almost perfectly diagonal, and precision, recall, and F1-scores reach unity for all classes. Detailed quantitative results, including the classification report and confusion matrix, are provided in Appendix B.

These results confirm that the selected architecture generalizes extremely well to unseen data drawn from the same underlying distribution.

5.2 External Out-of-Distribution Evaluation

Custom Dataset description

The custom dataset used for the out-of-distribution evaluation consists of a small collection of hand gesture images representing the three target classes (23 images for *rock* and *scissors*, 22 images for *paper*), acquired outside the original training distribution. The images were captured using consumer-grade devices under unconstrained conditions and include hands belonging to multiple individuals rather than a single subject. This choice introduces natural variability in hand shape, skin tone, finger proportions, and gesture execution style, thereby increasing the heterogeneity of the dataset.

In addition to subject variability, the images exhibit substantial diversity in background, lighting conditions, and viewpoint. Backgrounds range from uniform surfaces to textured environments such as wooden floors, tiled areas, and indoor furnishings, while lighting conditions vary from diffuse natural light to indoor artificial illumination. The hand poses are not strictly canonical: gestures may be partially rotated, cropped, or captured at non-standard distances and angles. Accessories such as watches or wristbands are also present in some images, further increasing visual complexity.

Although the dataset is approximately class-balanced, it is intentionally small and uncurated, reflecting realistic acquisition conditions rather than controlled laboratory settings. As such, it is not intended to provide statistically robust performance estimates, but rather to serve as a qualitative stress test for model robustness and generalization under domain shift. The dataset allows for a focused analysis of how well models trained on clean, curated data transfer to real-world scenarios characterized by higher intra-class variability and reduced visual consistency.

Out-of-Distribution Evaluation on Custom Data

The evaluation on the custom hand-gesture dataset highlights a clear and systematic difference in robustness across the three architectures under out-of-distribution conditions. Despite the dataset being nearly class-balanced, models A and B exhibit a pronounced bias towards the *scissors* class, as evidenced by highly skewed predicted class distributions and extreme mean predicted priors. In particular, model B collapses almost entirely onto *scissors* (predicted prior $\approx$ [0.003, 0.001, 0.996]), while model A shows a slightly milder but still substantial preference for the same class.

This behaviour is reflected in their confusion matrices and per-class metrics, where *scissors* achieves comparatively higher F1-scores than *rock* and especially *paper* for models A and B, with *paper* being strongly penalized (Table 5). This pattern indicates that, under domain shift, the simpler architectures systematically favour the most visually distinctive class, improving performance on *scissors* at the expense of degraded discrimination between *rock* and *paper*. The corresponding misclassified-image grids further reveal that visually ambiguous *rock* and *paper* gestures are consistently mapped to *scissors* with high confidence, supporting the interpretation of a bias-driven error structure rather than class-specific noise.

Table 5: *Per-class F1-score on the custom hand-gesture dataset.*

Model	F1 (rock)	F1 (paper)	F1 (scissors)
A	0.318	0.077	0.455
B	0.188	0.167	0.382
C	0.458	0.509	0.364

Model C, while not immune to domain shift, substantially mitigates this failure mode. In terms of top-1 decisions, it avoids degenerating into a single-class predictor and yields higher overall accuracy and macro F1-score, with a confusion matrix that preserves meaningful structure. Although the average predicted probability assigned to *scissors* is close to zero before recalibration (Appendix C, Table 8), the confusion matrix shows that the model still assigns *scissors* as the top-1 label for some samples. This indicates a calibration/score-shift effect, rather than a complete collapse in which the argmax always selects a single class.

These trends are summarized quantitatively by the macro-averaged evaluation metrics reported in Table 6 and are visually corroborated by the confusion matrices in Appendix C (Figure 7), which clearly illustrate the systematic class collapse observed for models A and B and its mitigation in model C.

Table 6: *Custom hand-gesture dataset evaluation by model. Metrics are macro-averaged over classes.*

Model	Accuracy	Macro Precision	Macro Recall	Macro F1
A	0.338	0.311	0.334	0.283
B	0.279	0.279	0.277	0.246
C	0.456	0.488	0.458	0.444

Overall, these results suggest that increased representational capacity and architectural regularization improve robustness to real-world variability, while also highlighting the intrinsic difficulty of reliably distinguishing hand gestures under uncontrolled acquisition conditions.

6 Conclusion and Future Work

This project investigated how increasing architectural complexity in convolutional neural networks affects both predictive performance and robustness to distributional shift in image classification tasks, using rock–paper–scissors hand gesture recognition as a controlled case study. Through a systematic comparison of three CNN architectures trained under identical optimization settings, the analysis demonstrated a clear relationship between model capacity and predictive performance. While the baseline architecture exhibited signs of underfitting and limited discriminative power, progressively deeper and better-regularized models achieved very strong in-distribution performance, confirming the benefits of increased representational expressiveness for this task.

Beyond in-distribution evaluation, the study placed particular emphasis on model robustness under domain shift by introducing an external test set composed of unconstrained, real-world hand gesture images. This generalization test revealed that simpler architectures tend to collapse toward visually dominant classes when confronted with unfamiliar acquisition conditions, leading to highly skewed predictions and poor class balance. In contrast, the most expressive model retained a more structured decision behaviour and substantially mitigated this failure mode, although some degradation in confidence calibration remained. These findings highlight that strong in-distribution validation performance alone is not sufficient to guarantee robustness in real-world scenarios and underscore the importance of explicitly evaluating models under out-of-distribution conditions.

Despite these encouraging results, several limitations must be acknowledged. The custom dataset used for the generalization test is intentionally small and qualitative in nature, preventing statistically robust performance estimates and limiting the strength of conclusions that can be drawn about generalization under domain shift. Moreover, all models were trained exclusively on a curated dataset, without the use of advanced data augmentation strategies or domain adaptation techniques that could further enhance robustness.

Future work could therefore explore the impact of more aggressive augmentation pipelines and the inclusion of real-world images during training to improve generalization under domain shift. Additionally, future experiments could evaluate the use of models pretrained on large image datasets to further enhance robustness in out-of-distribution settings. Investigating probabilistic calibration methods and confidence-aware decision mechanisms could also help address residual uncertainty when models are deployed in unconstrained environments.

Finally, in terms of potential applications, the final model could be deployed as a lightweight web application enabling interactive rock–paper–scissors play and real-time qualitative evaluation.

Appendix

A Additional Qualitative Diagnostics

A.1 Misclassified examples

Images are ranked by decreasing prediction confidence. Each panel reports the true label (T), the predicted label (P), and the associated softmax confidence score.

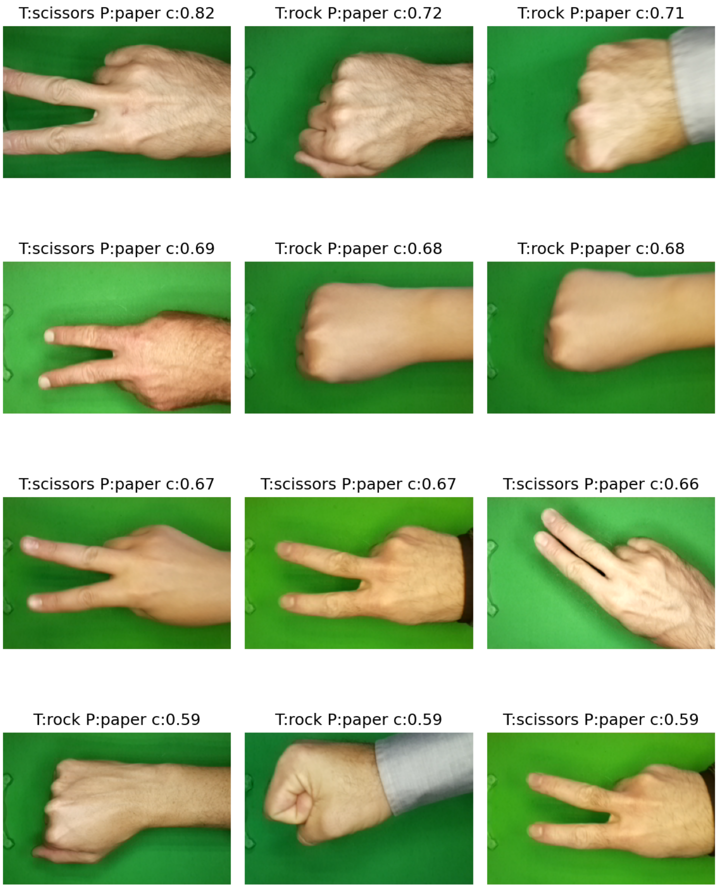

Figure 3: *Most confident misclassified validation samples for* ***Model A****.*

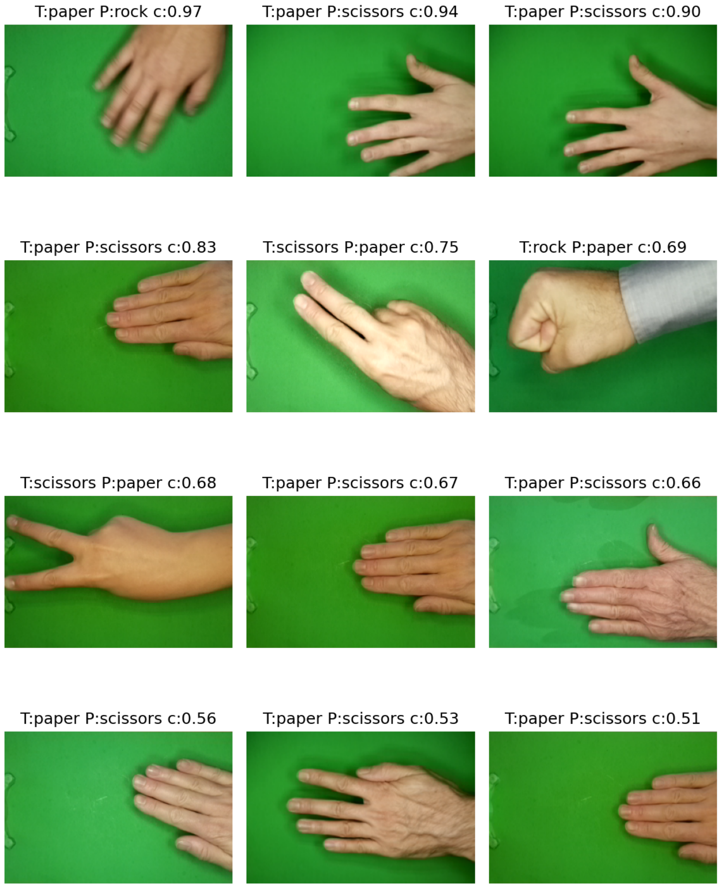

Figure 4: *Most confident misclassified validation samples for* ***Model B****.*

Figure 5: *Most confident misclassified validation samples for* ***Model C****.*

B External In-Distribution Evaluation Details

B.1 Classification report

Table 7: *Classification report for model C on the external in-distribution test set* rps-cv-images *(rounded values).*

Class	Precision	Recall	F1-score	Support
Rock	1.00	1.00	1.00	726
Paper	1.00	1.00	1.00	712
Scissors	1.00	1.00	1.00	750
Accuracy			1.00	2188
Macro avg	1.00	1.00	1.00	2188
Weighted avg	1.00	1.00	1.00	2188

B.2 Confusion matrix

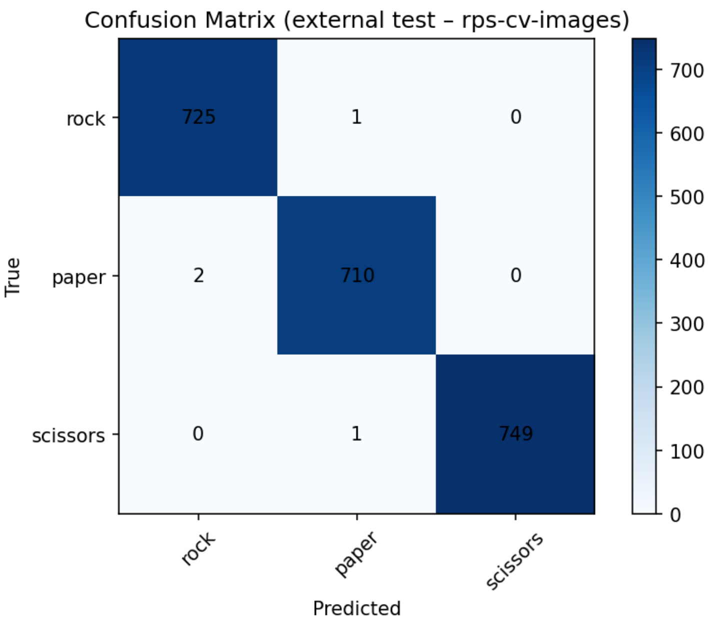

Figure 6: *Confusion matrix for model C on the external in-distribution test set (*rps-cv-images*).*

C External Out-Of-Distribution Evaluation Details

C.1 Confusion matrices

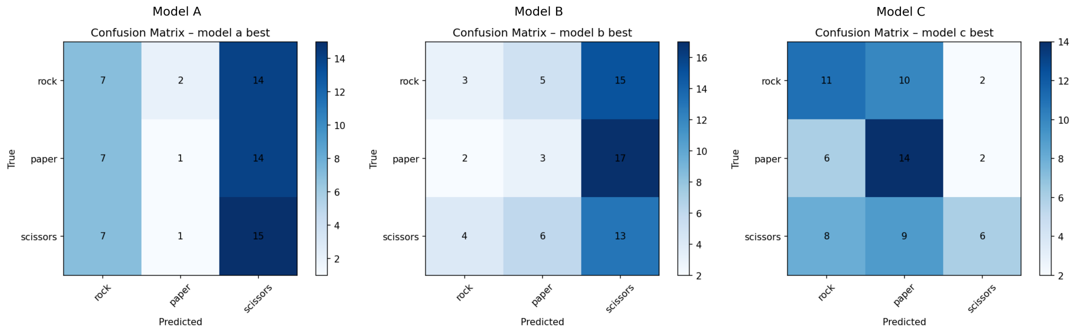

Figure 7: *Confusion matrices for the three architectures evaluated on the custom out-of-distribution dataset.*

C.2 Predicted class priors

Table 8: *Mean predicted class priors (pre-recalibration) for the three architectures on the custom out-of-distribution dataset.*

Model	**Rock**	**Paper**	**Scissors**
Model A	0.066	0.074	0.860
Model B	0.003	0.001	0.996
Model C	0.332	0.668	0.000
Target prior	0.333	0.333	0.333